

No. of Functional Features Included in the Solution

Date	July 9, 2026
Team ID	SWTID-2026-1262
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)
1	Predictive Crop Recommendation Engine	Maximizes production by processing seven soil/environmental values through an optimized Random Forest model to instantly output the most suitable crop match.	Backend Logic (train_model.py / app.py)	Done
2	Multi-Model Algorithmic Evaluator	Dynamically trains, evaluates, and prints accuracy metrics for Logistic Regression, KNN, Decision Tree, and Random Forest models to guarantee optimal selection thresholds.	Model Training (train_model.py)	Done
3	Interactive Scenario Tab Control Hub	Navigates cleanly between three user target states (Farmers, Researchers, Policymakers) within a responsive single-page architecture.	User Interface (index.html)	Done
4	Target Crop Suitability Auditor	Evaluates a user-specified crop against current climate values to flag environmental compatibility mismatches.	Backend Routing (app.py / result.html)	Done
5	Agro-Ecological Boundary Sanitizer	Intercepts incoming parameters and performs strict float type validation to protect against corrupt, invalid, or extreme entries.	Data Validation (app.py)	Done
6	Macro Policy & Efficiency Router	Dynamically pairs predictive metrics with pre-configured regional resource efficiency roadmaps and sustainable farming guidelines.	Decision Support Layer (app.py)	Done
7	Model Serialization Pipeline	Compiles, saves, and boots up model tree pathways natively from a local binary .pkl stream layout.	Database / Storage (model.pkl)	Done

8	Dynamic Result Summary Board	Displays input telemetry metrics alongside targeted agricultural roadmaps on a clean, scannable interface card.	User Interface (result.html)	Done
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Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5 (ML Model, Form Router, Recommendations, Validation, Persistence)
Additional / Nice-to-Have Features	3 (Scenario Tabs, Mismatch Audit, Policy Roadmap Layer)
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Responsive multi-scenario Bootstrap forms, dynamic result layout screens, error exception alert panels.
2	Backend / Logic	3	Random Forest prediction tree traversals, multi-algorithm accuracy rankers, boundary condition range parsing.
3	Database / Storage	2	Deserialized .pkl model streaming, in-memory Python dictionaries for sustainability context roadmaps.
4	API / Integration	0	Not applicable <i>(The system is intentionally self-contained locally on the local server workspace for maximum processing speed).</i>
5	Security / Authentication	1	Safe backend variable sanitization preventing type overrides and boundary exceptions.